



Loneliness, Depression, and Subjective Memory Complaints in Middle-Aged and Older Adults: The Role of Meaning in Life

Seung Chul Lee¹, Sungkun Cho¹, Yerin Shim¹, Dae-Hyun Yun², Taehyun Kim³, and Soo Rim Noh¹ ✉

¹Department of Psychology, Chungnam National University, Daejeon, Republic of Korea

²Department of Sport Science, Chungnam National University, Daejeon, Republic of Korea

³Department of Psychiatry, E-Lum Hospital, Bucheon, Republic of Korea

Objective This study examined whether depressive symptoms mediate the association between loneliness and subjective memory complaints (SMCs), and whether meaning in life (MIL) moderates this pathway, among middle-aged and older Korean adults.

Methods A cross-sectional survey was conducted with 580 adults aged 41–71 years (mean age=58.80; 54.8% males). Participants completed measures of loneliness, depression, SMCs, and MIL. Mediation (PROCESS Model 4) and moderated mediation (Model 7) analyses were performed using 5,000 bootstrap resamples while adjusting for demographic covariates. Supplementary analyses examined MLQ subscales (Presence and Search for Meaning) and age-stratified models.

Results Loneliness was significantly associated with depressive symptoms ($\beta=0.523$, $p<0.001$) and SMCs ($\beta=0.131$, $p=0.008$). Depression was also associated with SMCs ($\beta=0.200$, $p<0.001$) and partially mediated the loneliness–SMCs link (indirect effect=0.011; 95% confidence interval [CI] [0.006, 0.018]). The index of moderated mediation was significant (index=-0.0002, 95% CI [-0.0004, -0.0001]). MIL moderated the loneliness–depression association (interaction $B=-0.005$, $p<0.001$), with a stronger slope at low MIL ($B=0.237$) than at high MIL ($B=0.148$). Subscale analyses showed similar patterns for presence and search when tested separately, but neither showed a unique moderating effect when entered simultaneously. Age-stratified analyses showed depression mediated the loneliness–SMCs link only in older adults, whereas MIL attenuated the loneliness–depression association only in middle-aged adults.

Conclusion Loneliness was associated with SMCs partly through depressive symptoms, and MIL may buffer this pathway. These mechanisms differed by life stage, with stronger evidence for depression as a mediator in later life than in midlife, suggesting age-tailored approaches targeting emotional distress and life purpose.

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Keywords Subjective memory complaints; Loneliness; Depression; Meaning in life; Aging.

INTRODUCTION

With increasing life expectancies, cognitive impairment, including dementia, has become a critical health issue across age groups. Subjective cognitive decline, often observed through subjective memory complaints (SMCs), is now recognized as an early indicator of cognitive deterioration. SMCs refer to an individual's self-perception of memory deterioration without objectively measurable deficits and can serve as a precursor to more severe cognitive conditions such as mild cognitive im-

pairment (MCI) and Alzheimer's disease (AD).^{1,2} Studies have shown that older adults with SMCs face a higher risk of future cognitive decline.³⁻⁵

The prevalence of SMCs varies across populations. In the United States, the prevalence stands at 10.1% among adults aged 45–64 and 11.7% for those 65 years and older.⁶ In contrast, South Korea has significantly higher reported rates, with 17.4% of middle-aged adults and 29.4% of older adults experiencing SMCs.⁷ This increase parallels a marked rise in dementia incidence in South Korea, where rates increased substantially between 2004 and 2015, posing a pressing public health challenge.^{8,9} Given these elevated prevalence rates—particularly among middle-aged adults—understanding the psychological mechanisms associated with SMCs in this transitional period becomes especially important.

Against this background, research on SMCs has traditionally focused on older adults. However, emerging evidence suggests that middle-aged adults represent a critical transitional

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✉ **Correspondence:** Soo Rim Noh, PhD

Department of Psychology, Chungnam National University, 99 Daehak-ro, Yuseong-gu, Daejeon 34134, Republic of Korea

Tel: +82-42-821-6365, **E-mail:** srnoh@cnu.ac.kr

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group in the trajectory of cognitive aging. In this population, early signs of cognitive vulnerability may first emerge, and interventions may yield the greatest preventive benefits.^{10,11} Despite this, the mechanisms underlying SMCs in midlife remain poorly understood. Among the psychological factors implicated in SMCs, loneliness and depression have received growing attention. Of these, loneliness may be especially relevant because it functions as a chronic psychosocial stressor that can uniquely compromise both affective and cognitive functioning.¹² Although both have been independently linked to memory complaints, fewer studies have investigated how these factors may interact to affect cognitive self-appraisal—particularly in non-Western contexts.¹³ Loneliness, defined as a subjective and distressing experience arising from a mismatch between desired and actual social relationships, is conceptually distinct from objective social isolation and has been increasingly recognized as a potent psychological stressor.^{14,15} It is not merely associated with depression but has been shown to contribute to its development through both physiological and psychological mechanisms. For instance, chronic loneliness may trigger persistent stress responses that negatively affect neuroendocrine and immune function, heightening emotional vulnerability and depleting affective resources. These processes may foster feelings of hopelessness, worthlessness, and fatigue, all of which are characteristic of depressive states.¹⁶ In addition, loneliness appears to compromise attentional control by promoting hypervigilance for social threats, thereby reducing working memory capacity and increasing susceptibility to memory complaints.^{12,17}

Depression, in turn, has consistently been associated with SMCs—even in the absence of measurable cognitive decline—and may influence memory self-appraisal through various cognitive and affective pathways.^{18,19} The negative cognitive triad,²⁰ overgeneral autobiographical memory (OGM), and rumination all appear to maintain maladaptive self-beliefs about cognitive functioning. OGM restricts access to specific personal memories, weakening retrieval of success-related information and reinforcing perceived decline.^{21,22} Rumination sustains focus on past failures and inhibits disengagement from negative affect, which may consume cognitive resources and amplify subjective concerns about memory.²³ Moreover, depressive affect may reduce available cognitive capacity and increase the tendency to misattribute minor attentional lapses to more serious cognitive impairment.²⁴ Collectively, these mechanisms provide a theoretical rationale for conceptualizing depression as a mediator in the link between loneliness and SMCs, particularly among midlife and early-old-age adults who may be more susceptible to psychosocial stress and memory-related concerns. However, few studies have examined these interrelationships in an integrated framework, particu-

larly in non-Western populations.

Meaning in life (MIL), defined as an individual's sense of purpose and coherence in life, has been proposed as a potential protective factor in the context of loneliness and depression.²⁵ Rooted in Frankl's existential theory, MIL represents a fundamental human drive; its absence is associated with affective distress and disengagement.²⁶ Empirical work consistently demonstrates that individuals with stronger MIL report lower loneliness, fewer depressive symptoms, and reduced memory complaints.^{27,28} A large European study of older adults further indicated that MIL attenuated the indirect association between loneliness and depression, accounting for a substantial proportion of this link.²⁹ Although MIL has been conceptualized as both a mediator and a moderator in previous studies, the present study focuses on its moderating role, based on the rationale that MIL may alter the strength—rather than transmit the effect—of the association between loneliness and depression. Beyond mediation, emerging evidence suggests that MIL moderates the loneliness–depression pathway by buffering emotional consequences of social disconnection. For example, older adults with high MIL exhibited smaller increases in depressive symptoms after major stressors,³⁰ and greater life purpose mitigated the adverse effects of thwarted belongingness on psychological well-being.³¹ In collectivistic contexts, meaning is often derived from social roles and relational obligations; under such conditions, MIL may partly compensate for unmet belongingness needs. The Meaning in Life Questionnaire (MLQ) distinguishes presence of meaning from search for meaning; prior work suggests that presence primarily drives the buffering effect, whereas search may show neutral or even positive associations with distress.³² MIL is also linked to cognitive resilience through improved attentional control and stress regulation.³² Longitudinal analyses reveal that a strong sense of purpose predicts a lower risk and delayed onset of cognitive impairment across diverse groups.³³ Taken together, these findings support MIL as a psychological resource that may buffer the impact of loneliness on depressive affect and, in turn, mitigate downstream cognitive consequences.^{34,35}

Building on prior research linking loneliness, depression, and SMCs, the present study examines whether depression mediates the association between loneliness and SMCs. We hypothesize that loneliness affects SMCs both directly and indirectly through its impact on depressive symptoms. Furthermore, we propose that MIL moderates this indirect pathway by attenuating the effect of loneliness on depression. In other words, individuals with higher levels of MIL may be less susceptible to emotional consequences of loneliness, thereby reducing its downstream impact on perceived memory decline. To test this conceptual model, we employed a moderated

mediation framework. Figure 1 illustrates the proposed model.

METHODS

Sample

The study surveyed 694 middle-aged and older adults aged 41–71 years, recruited through the Media Real Research Korea panel using a stratified random sampling approach to ensure representation of the general Korean population. Eligibility was determined through five screening questions administered during the panel recruitment process. Specifically, individuals were excluded if they reported 1) a history of head injury or brain-related disorders; 2) a current or past diagnosis of a psychiatric disorder (e.g., depression); 3) treatment for depression within the past year, including medication, psychotherapy or counseling, hospitalization, or other clinical care; 4) a diagnosis of dementia or MCI; or 5) a history of treatment for substance or alcohol addiction. For depression, dementia, and MCI, diagnosis alone was sufficient for exclusion, whereas the treatment criterion referred only to treatment received within the past year. These screening items were administered by the survey panel company during initial participant selection, and all respondents in the analytic dataset reported no exclusion conditions at screening, as confirmed by raw-data review prior to further cleaning.

Of the 694 initial respondents, 114 (16.4%) were excluded due to low-quality responses, defined as a survey completion time of less than 20 minutes ($n=72$) or straight-lining for 10 or more consecutive items, primarily in scales containing reverse-scored items ($n=42$). These cases were automatically flagged by the survey company and subsequently verified by the research team. The final analytic sample included 580 participants ($M_{\text{age}}=58.80$, standard deviation [SD]=6.61 years; 54.8% male), who completed the online survey in February 2022. This study was approved by the Institutional Review Board of Chungnam National University (IRB No. 202112-SB-252-01), and all participants provided informed consent. Demographic characteristics of the analytic sample are presented in Table 1, and a comparison of the excluded and retained groups is provided in Supplementary Table 1.

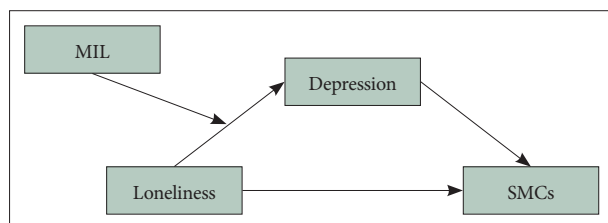


Figure 1. Hypothesized moderated mediation model of the effect of loneliness on subjective memory complaints (SMCs) through depression, moderated by meaning in life (MIL).

Table 1. Demographic characteristics of the participants (N=580)

Characteristics	N (%)
Female sex	262 (45.20)
Age (yr)	
41–49	52 (8.97)
50–59	222 (38.28)
60+	306 (52.76)
Education	
Elementary school	11 (1.90)
Middle school	21 (3.62)
High school	255 (43.97)
College	242 (41.72)
Greater than college	51 (8.79)
Marital status	
Married/living with a partner	470 (81.03)
Never married	29 (5.00)
Separated/divorced	66 (11.38)
Widowed	15 (2.59)
Monthly income	
Less than \$1,540	150 (25.86)
\$1,540–\$2,310	140 (24.14)
\$2,311–\$3,080	121 (20.86)
\$3,081–\$3,850	78 (13.45)
\$3,851–\$4,620	40 (6.90)
More than \$4,620	51 (8.79)
Household size	
Living alone	41 (7.07)
Two-person household	281 (48.45)
Three or more-person household	258 (44.48)
Number of chronic diseases	
0	299 (51.55)
1	94 (16.21)
2+	187 (32.24)
Participation in religious activities	
Yes	294 (50.69)
No	286 (49.31)

Covariates included in the main analysis were coded as follows: age was treated as a continuous variable; education was coded into four levels: elementary=1, middle school=2, high school=3, and college or higher=4; marital status was categorized as married/living with a partner=1, never married=2, separated/divorced=2, and widowed=2; monthly income was grouped into three categories: less than \$2,310=1, \$2,311–\$4,620=2, and more than \$4,620=3; household size was coded as living alone=1, two-person household=2, and three or more people=3; religious participation was coded as yes=1 and no=0; chronic disease count was categorized as 0=0, 1=1, and 2 or more=2.

Measures

Subjective Memory Complaints Questionnaire

SMCs were measured using four binary-response items from the Subjective Memory Complaints Questionnaire³⁶: “Do you think that you have a memory problem?”, “Is your memory worse than 10 years ago?”, “Is your memory poorer than that of others your age?”, and “Does memory decline affect your daily life?”. Each item was scored as no (0) or yes (1), and scores were summed (range: 0–4), with higher values indicating greater memory complaints. The four dichotomous items showed modest internal consistency based on conventional indices (Cronbach’s $\alpha=0.668$, McDonald’s $\omega=0.679$). Given the binary response format, we additionally estimated tetrachoric-based ordinal reliability,³⁷ which was high (ordinal $\alpha=0.908$) (Supplementary Table 2).

UCLA Loneliness Scale

Loneliness was assessed using the Korean version of the UCLA Loneliness Scale,³⁸ which included 20 items rated on a 4-point Likert scale (0=never, 3=always). The scale includes two subscales: social isolation (11 items) and social connectedness (9 reverse-scored items). Example items include “I feel completely alone” and “I am unhappy being so withdrawn.” Higher total scores reflect greater levels of perceived loneliness. The internal consistency for this sample was high (Cronbach’s $\alpha=0.875$).

Center for Epidemiologic Studies Depression Scale

Depressive symptoms were measured with the 10-item Center for Epidemiologic Studies Depression Scale,³⁹ validated in Korea.⁴⁰ Participants rated items such as “I felt depressed” and “I had trouble keeping my mind on what I was doing” on a 4-point Likert scale (0=rarely or none of the time, 1=some of the time, 2=occasionally, 3=most or all of the time). Scores were summed (range: 0–30), with higher scores indicating more depressive symptoms. Cronbach’s α for the scale in this study was 0.833.

MLQ

The MLQ²⁵ adapted into Korean by Won et al.,⁴¹ was used to assess participants’ sense of life meaning. The MLQ consists of 10 items measuring both the presence of meaning (5 items) and the search for meaning (5 items), rated on a 7-point Likert scale (1=absolutely untrue to 7=absolutely true). Example items include “I understand my life’s meaning” and “I am always looking to find my life’s purpose.” Higher scores indicate greater levels of life meaning. Internal consistency was excellent (Cronbach’s $\alpha=0.920$). Although the MLQ was originally validated with university students,⁴¹ it has also been

used in Korean middle-aged and older adult samples, with acceptable psychometric properties reported in these age groups.^{42,43}

Statistical analysis

We used Hayes’ PROCESS macro⁴⁴ (Models 4 and 7) in SPSS 26.0 (IBM Corp.) to test the proposed mediation and moderated mediation models. Model 4 was used to examine the indirect effect of loneliness on SMCs through depression, and Model 7 was used to test whether MIL moderated this pathway. Indirect effects were evaluated using 5,000 bootstrap resamples and 95% confidence intervals (CIs); effects were considered significant when the CI did not include zero. For the moderated mediation, simple slopes were probed at the mean of MIL and at ± 1 SD. Loneliness and MIL were mean-centered to reduce multicollinearity.⁴⁵

Before model testing, descriptive statistics and Pearson correlation coefficients were calculated for all key variables. Sex, age, education, income, marital status, household size, participation in religious activities, and chronic diseases were included as covariates based on previous literature.^{13,46,47} As an additional sensitivity analysis, we re-estimated the primary mediation and moderated mediation models in the full sample ($n=694$), including the 114 cases excluded from the main analyses due to response-quality concerns. These models used the same covariates and bootstrap procedures as the primary analyses and were examined to determine whether the overall pattern of findings was materially altered by the exclusions.

To evaluate the robustness of the primary findings and examine potential heterogeneity, we conducted two supplementary analyses in addition to the primary models using the total MLQ score as the moderator. The total MLQ score was retained as the primary moderator because the present study focused on the overall buffering role of MIL at the construct level.²⁵ First, we re-estimated the moderated mediation model separately for presence of meaning and search for meaning, and then entered both subscales simultaneously to examine their unique contributions, given prior evidence that the two dimensions may relate differently to psychological distress^{48,49}; overlap between the subscales was assessed using bivariate correlation and variance inflation factors (VIFs).⁵⁰ Second, because the sample spanned midlife and older adulthood (41–71 years), we conducted stratified analyses for middle-aged adults (41–59 years) and older adults (60+ years), consistent with prior work showing age-related differences in the structure and correlates of MIL.⁵¹ All supplementary models included the same covariates, used 5,000 bootstrap resamples to derive 95% CIs, and mean-centered predictors prior to creating interaction terms.

RESULTS

Participant characteristics

Table 1 presents the demographic characteristics of the participants. Approximately half were aged 41–59 years and the other half were 60 years or older. Most participants had completed high school or college and were married or living with a partner. Monthly household income showed substantial variability. Regarding health status, over half reported no chronic diseases, while the remainder had one or more chronic conditions. Participation in religious activities was evenly divided.

Correlations and descriptive statistics

Table 2 displays the descriptive statistics and Pearson correlation coefficients among the four key variables: SMCs, loneliness, depression, and MIL. On average, participants reported low-to-moderate levels of SMCs ($M=1.18$, $SD=1.10$). SMCs were significantly positively correlated with loneliness ($r=0.23$, $p<0.001$) and depression ($r=0.27$, $p<0.001$). Loneliness and depression were also strongly positively correlated ($r=0.56$, $p<0.001$). In contrast, MIL was negatively correlated with all

three variables: SMCs ($r=-0.22$, $p<0.001$), loneliness ($r=-0.47$, $p<0.001$), and depression ($r=-0.38$, $p<0.001$).

Mediation analysis

To examine whether depression mediates the association between loneliness and SMCs, we conducted a mediation analysis using PROCESS Model 4.⁴⁴ The total effect of loneliness on SMCs was statistically significant ($\beta=0.236$, $p<0.001$). After accounting for depression, the direct effect of loneliness on SMCs remained significant ($\beta=0.131$, $p=0.008$), as did the direct effect of depression on SMCs ($\beta=0.200$, $p<0.001$). The direct effect of loneliness on depression was also significant ($\beta=0.523$, $p<0.001$). As shown in Table 3, the indirect effect of loneliness on SMCs via depression was statistically significant (effect=0.011, boot standard error=0.003, 95% CI [0.006, 0.018]). Comparable results were obtained in the unadjusted model, with no meaningful changes in the direction, magnitude, or significance of the direct and indirect effects.

Moderated mediation analysis

We further explored whether MIL moderates the indirect effect of loneliness on SMCs via depression using PROCESS Model 7.⁴⁴ Table 4 shows that the interaction between loneliness and MIL was significant in predicting depression ($B=-0.005$, $p<0.001$), indicating that MIL moderated the association between loneliness and depression. Specifically, the effect of loneliness on depression was stronger at low levels of MIL ($M-1$ SD: $B=0.237$, $p<0.001$) compared to high levels ($M+1$ SD: $B=0.148$, $p<0.001$). As illustrated in Figure 2, the association between loneliness and depression was stronger when MIL was lower and weaker when MIL was higher.

The moderated mediation model also showed that the conditional indirect effects of loneliness on SMCs through depression remained significant at both low and high levels of MIL. At low MIL ($M-1$ SD), the conditional indirect effect was

Table 2. Pearson correlation coefficients and descriptive statistics for major variables (N=580)

	1	2	3	4
1. SMCs	1			
2. Loneliness	0.23***	1		
3. Depression	0.27***	0.56***	1	
4. MIL	-0.22***	-0.47***	-0.38***	1
Mean	1.18	18.98	5.78	50.46
SD	1.10	10.14	4.22	9.72
Skewness	0.97	0.28	1.60	-0.32
Kurtosis	0.36	0.02	4.53	0.26

*** $p<0.001$. SMCs, subjective memory complaints; MIL, meaning in life; SD, standard deviation.

Table 3. Mediation (Model 4) analysis results (N=580)

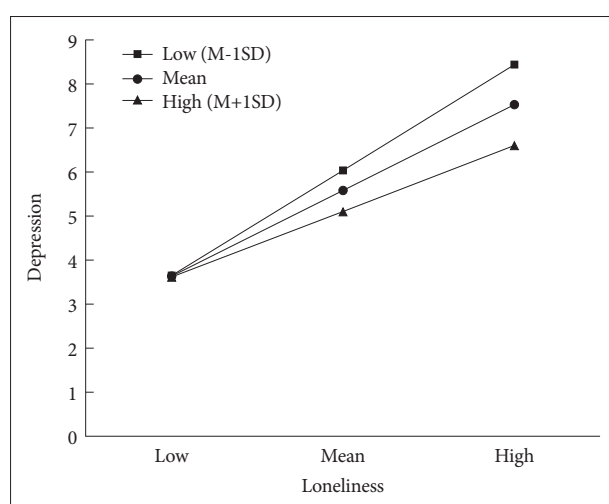
Effect	B (SE)	β	t	p
Total effect				
Loneliness $\rightarrow$ SMCs	0.026 (0.005)	0.236	5.592	<0.001
Direct effect				
Loneliness $\rightarrow$ SMCs	0.014 (0.005)	0.131	2.685	0.008
Path				
Loneliness $\rightarrow$ Depression	0.218 (0.015)	0.523	14.625	<0.001
Depression $\rightarrow$ SMCs	0.052 (0.013)	0.200	4.110	<0.001
	B	β	Boot SE	Boot 95% CI
Indirect effect				
Loneliness $\rightarrow$ Depression $\rightarrow$ SMCs	0.011	0.105	0.003	0.006 to 0.018

Bootstrap sample size=5,000. SE, standard error; Boot, bootstrap; CI, confidence interval; SMCs, subjective memory complaints.

Table 4. Moderated mediation (Model 7) analysis results (N=580)

Effect	B	SE	t	p
Direct effect				
Loneliness → SMCs	0.014	0.005	2.685	0.008
Path				
Loneliness → Depression	0.192	0.016	11.819	<0.001
Interaction → Depression	-0.005	0.001	-3.876	<0.001
Depression → SMCs	0.052	0.013	4.110	<0.001
Indirect effect	B	Boot SE	Boot 95% CI	
Loneliness → Depression → SMCs				
MIL low (-1 SD)	0.012	0.003	0.006 to 0.019	
MIL high (+1 SD)	0.008	0.002	0.004 to 0.013	
Index of moderated mediation	-0.0002	0.0001	-0.0004 to -0.0001	

Bootstrap sample size=5,000. Loneliness and MIL were mean-centered. SE, standard error; Boot, bootstrap; CI, confidence interval; SMCs, subjective memory complaints; MIL, meaning in life; SD, standard deviation.

**Figure 2.** Moderating effect of meaning in life on the relationship between loneliness and depression. SD, standard deviation.

0.012 (95% CI [0.006, 0.019]); at high MIL (M+1 SD), it was 0.008 (95% CI [0.004, 0.013]). The index of moderated mediation was -0.0002 (95% CI [-0.0004, -0.0001]), indicating a statistically significant but small moderation effect. The unadjusted model yielded materially similar results, with no meaningful changes in the direction, magnitude, or significance of the interaction term, conditional indirect effects, or index of moderated mediation.

To aid interpretation of the modest interaction effect, we conducted additional post hoc sensitivity and power analyses for the focal loneliness×MIL interaction term predicting depressive symptoms. The interaction term accounted for an additional 1.67% of variance in depression, $\Delta R^2=0.017$, $F(1, 567)=15.02$, $p<0.001$, corresponding to a small effect size (Cohen's $f^2=0.027$).⁵² Sensitivity analyses indicated that the present sample was sufficient to detect small interaction effects;

detailed results are presented in Supplementary Table 3.

Sensitivity analysis including excluded cases

To examine whether the exclusion of low-quality responses materially affected the findings, we re-estimated the primary mediation and moderated mediation models in the full sample ($n=694$), including the 114 excluded cases. The pattern of results remained materially unchanged. In the mediation model, loneliness continued to show significant total and direct effects on SMCs (total effect: $B=0.026$, $p<0.001$; direct effect: $B=0.014$, $p=0.008$), and the indirect effect through depression remained significant ($B=0.013$, 95% CI [0.007, 0.019]). In the moderated mediation model, the loneliness×MIL interaction remained significant in predicting depression ($B=-0.005$, $p<0.001$), the conditional indirect effects remained significant at both low and high levels of MIL (low: $B=0.014$, 95% CI [0.008, 0.020]; high: $B=0.009$, 95% CI [0.005, 0.013]), and the index of moderated mediation also remained significant ($B=-0.0003$, 95% CI [-0.0004, -0.0001]). These findings indicate that the main results were robust to the inclusion of cases excluded for response-quality concerns. Detailed results are presented in Supplementary Table 4.

Supplementary analyses using MLQ subscales

The two subscales were strongly correlated ($r=0.79$), and VIFs ranged from 2.77 to 3.11, remaining below commonly used thresholds for problematic multicollinearity.⁵⁰

We then repeated PROCESS Model 7 separately for each subscale. When entered individually, both presence of meaning ($B=-0.008$, $p<0.001$; index=-0.0004, 95% CI [-0.0008, -0.0001]) and search for meaning ($B=-0.009$, $p<0.001$; index=-0.0004, 95% CI [-0.0008, -0.0001]) significantly moderated the association between loneliness and depression. In

both models, the conditional indirect effect of loneliness on SMCs through depression decreased as the moderator increased (presence: from 0.012 to 0.007; search: from 0.013 to 0.009) (Supplementary Table 5).

However, when presence and search were entered simultaneously in a combined PROCESS Model 9, neither interaction term was statistically significant (loneliness $\times$ presence: $B=-0.007$, $p=0.088$; loneliness $\times$ search: $B=-0.003$, $p=0.544$), and detailed results are presented in Supplementary Table 6.

Supplementary stratified analyses by age group

Given the broad age range of the sample, we conducted supplementary stratified analyses by estimating the mediation (Model 4) and moderated mediation (Model 7) models separately in middle-aged adults (41–59 years) and older adults (60+ years). Preliminary comparisons indicated that, among the key variables, only depression differed significantly between the two age groups, with middle-aged adults reporting slightly higher depressive symptoms ($M=6.20$, $SD=4.53$) than older adults ($M=5.40$, $SD=3.89$, $t(578)=2.30$, $p=0.022$) (Supplementary Table 7).

In the middle-aged subgroup, loneliness showed a significant total effect on SMCs ($\beta=0.178$, $p=0.004$). However, the direct effect of loneliness on SMCs was not significant after accounting for depression ($\beta=0.108$, $p=0.135$). In addition, neither the indirect effect via depression (effect=0.007, 95% CI [-0.001, 0.016]) nor the index of moderated mediation (index=-0.0002, 95% CI [-0.0005, 0.0000]) was statistically significant, although the loneliness $\times$ MIL interaction significantly predicted depression ($B=-0.007$, $p<0.001$) (Supplementary Table 8).

In the older subgroup, both the direct effect ($\beta=0.162$, $p=0.016$) and indirect effect (effect=0.016, 95% CI [0.008, 0.026]) of loneliness on SMCs were significant, whereas the loneliness $\times$ MIL interaction was not ($B=0.000$, $p=0.898$), and the index of moderated mediation was also not significant (index=0.0000, 95% CI [-0.0003, 0.0003]) (Supplementary Table 8).

DISCUSSION

This study builds on existing research by examining the mediating role of depression in the relationship between loneliness and SMCs, and the moderating role of MIL in middle-aged and older Korean adults. Prior studies have demonstrated that loneliness significantly predicts depressive symptoms, which in turn contributes to cognitive vulnerability, including SMCs.^{12,18} Our findings align with this evidence and suggest that depressive symptoms may partly account for the association between loneliness and self-perceived cognitive decline.¹⁹

The current findings suggest that depression may function

as a critical pathway between loneliness and SMCs. This is consistent with twin studies in which SMCs remained significantly associated with depressive symptoms even after controlling for genetic and family environmental factors.⁵³ Longitudinal analyses likewise reveal that perceived memory decline predicts subsequent depressive symptoms and that concurrent associations between depressive symptoms and memory complaints are strong; however, the reverse longitudinal relationship—depressive symptoms predicting later memory complaints—has been less consistently observed.⁴⁶ These findings collectively suggest that depressive symptoms may represent an important pathway linking loneliness to memory complaints.

Consistent with prior theoretical accounts, depressive symptoms may intensify SMCs through a combination of cognitive and affective distortions. Specifically, individuals with depression tend to engage in ruminative thought patterns and retrieve OGMs, both of which may impair memory self-appraisal by limiting access to specific positive episodic recall and prolonging negative emotional cycles.^{22,23} These mechanisms may help explain why individuals with depressive symptoms tend to interpret ordinary attentional lapses as signs of cognitive decline, giving rise to SMCs even in the absence of objective impairment. In addition to these cognitive processes, structural and functional alterations in memory-monitoring regions—such as the hippocampus and prefrontal cortex—may further contribute to distorted perceptions of memory performance.⁵⁴ Taken together, these pathways provide a plausible explanation for how emotional distress, particularly when triggered by loneliness, may culminate in elevated SMCs via depressive symptomatology.

This study adds to existing evidence by suggesting that MIL may mitigate the link between loneliness and depression, and in turn, its indirect association with SMCs. Previous research has recognized MIL as a psychological resource that enhances resilience in the face of emotional adversity.^{27,55} Our findings suggest that individuals with higher MIL reported fewer depressive symptoms in the context of loneliness, and in turn were less likely to report memory-related complaints. This aligns with theoretical frameworks that view purpose in life as a cognitive-emotional resource that regulates affect and strengthens coping.^{56,57} This cultural extension is especially meaningful, as most prior research has been conducted in Western samples, leaving pathways underexamined in East Asian contexts. This interpretation is also consistent with findings during the COVID-19 pandemic showing that individuals with a stronger sense of MIL reported lower loneliness through adaptive coping strategies.²⁸

Supplementary analyses of the MLQ subscales warrant a more nuanced interpretation of the total-score findings ob-

tained with the MLQ.^{25,41} Prior research suggests that presence of meaning and search for meaning may have different psychological implications, with presence more consistently linked to well-being and search sometimes showing more mixed associations with distress^{48,49,58}; this distinction may be especially relevant in midlife and older adulthood, when meaning is often expressed less as active searching and more as integrating life experiences, maintaining continuity, and consolidating a sense of coherence and legacy.⁵¹ In the present study, both Presence and Search showed a similar buffering pattern when tested separately, but neither showed a unique moderating effect when entered simultaneously. Given the strong correlation between the two subscales ($r=0.79$), the buffering effect observed for the total MIL score may primarily reflect variance shared across the two dimensions rather than clearly separable subscale-specific mechanisms. Accordingly, although the distinction between presence and search remains theoretically meaningful, subscale-specific interpretations should be made cautiously in the present sample.⁵⁹

These findings have practical implications for psychological interventions targeting cognitive well-being in aging populations. First, the mediating role of depression suggests that depressive symptoms may represent an important intervention target in the association between loneliness and SMCs. From a clinical and public health perspective, screening for depressive symptoms among lonely middle-aged and older adults may therefore be relevant in primary care or community mental health settings. Previous research has often examined loneliness and depression separately.⁶⁰ Our findings instead underscore the relevance of more integrative approaches that address both affective and cognitive domains. Second, the buffering role of MIL points to the potential value of meaning-centered interventions. Meaning-centered group therapy, logotherapy-based approaches, and life-review interventions have been shown to enhance purpose and reduce emotional distress in older adults, suggesting their relevance for aging populations vulnerable to both emotional and cognitive distress.^{26,61,62} Our age-stratified findings further suggest that these interventions may be more effective when tailored to developmental needs. In middle-aged adults, fostering goal-directed purpose, future-oriented engagement, and social connectedness may strengthen resilience against loneliness-related distress. In older adults, by contrast, life-review and legacy-oriented approaches may be better suited to the challenges of late-life meaning-making and role transition. Evidence that individuals with a strong sense of purpose report less perceived cognitive decline in midlife,³² and that higher purpose in life predicts a lower risk and delayed onset of cognitive impairment in a population-based Health and Retirement Study (HRS) cohort,³³ further supports the relevance of meaning-

making for cognitive resilience. Third, given the high prevalence of SMCs in Korea and their associations with loneliness and depression, community-based programs that promote social connection, emotional support, and a sense of purpose may be relevant for prevention-oriented mental health efforts in middle-aged and older adults.⁷

The age-stratified analyses suggested that the associations among loneliness, depression, and SMCs may not be uniform across middle-aged (41–59 years) and older (60+ years) adults (Supplementary Tables 7 and 8). Although middle-aged adults reported slightly higher depressive symptoms than older adults, the main difference lay in the pattern of associations: in midlife, MIL significantly attenuated the association between loneliness and depression, whereas the indirect effect of loneliness on SMCs through depression was not statistically robust. By contrast, in older adults, depression emerged as a more consistent mediator linking loneliness to SMCs, whereas MIL did not significantly moderate the association between loneliness and depression. One interpretation is that, in midlife, MIL may function more as a general emotional resilience resource than as a specific buffer against loneliness-related memory complaints, whereas in later life the loneliness–depression relationship may be less amenable to buffering by psychological resources alone.^{12,30} Collectively, these findings underscore the importance of developmentally informed models of cognitive vulnerability.

Although the interaction between loneliness and MIL was statistically significant in the full sample, its magnitude was modest. The conditional indirect effect decreased from 0.012 at low MIL to 0.008 at high MIL, indicating an approximately 33% attenuation of the indirect pathway. The index of moderated mediation was small in absolute terms. However, this index depends on the scaling of the component variables,⁶³ which limits direct magnitude comparisons across studies. This pattern is nevertheless consistent with prior studies suggesting that psychological resources can exert modest but meaningful buffering effects.^{32,34,35} Although the observed moderation effect was small, supplementary sensitivity analyses indicated that the present sample was sufficient to detect interaction effects of this magnitude. Accordingly, the buffering role of MIL should be interpreted as statistically reliable but modest in magnitude (Supplementary Table 3). At the same time, MIL may function not only as a moderator but also as a mediator or downstream correlate shaped by loneliness and depression.^{29,64} Future studies should therefore test alternative models using longitudinal and cross-lagged designs, particularly given emerging evidence of bidirectional associations between MIL and loneliness/depression in older adults.⁶⁵

Several limitations should be considered. First, the cross-sectional design precludes causal inferences. Although medi-

ation and moderation analyses suggest directional patterns, longitudinal designs are needed to establish temporal sequences. Second, the study focused on Korean middle-aged and older adults, limiting generalizability to other age groups or cultural contexts. Future research should examine whether MIL plays a similar role in diverse populations. Third, all key variables were assessed through self-report in a single online survey, raising the possibility of common method bias and inflated associations among constructs.⁶⁶ Future studies should incorporate objective cognitive assessments, informant reports, or temporally separated measurements. Fourth, the exclusion of 114 respondents (16.4%) due to low-quality responses may have introduced selection bias. Excluded respondents differed from the analytic sample on several demographic and psychological characteristics (Supplementary Table 1), with somewhat higher loneliness and lower MIL in the excluded group. At the same time, sensitivity analyses re-estimating the primary models in the full sample including these excluded cases yielded a materially unchanged pattern of results (Supplementary Table 4), suggesting that the exclusions did not substantially alter the primary conclusions. Nonetheless, because many excluded cases were identified on the basis of clear response-quality concerns, their inclusion may also have introduced additional measurement error, and the possibility of some bias in representativeness cannot be fully ruled out. Finally, the four-item binary SMC measure showed only modest internal consistency based on conventional indices. Because dichotomous response formats can suppress variability and attenuate Pearson-based reliability estimates, we additionally examined tetrachoric-based ordinal reliability and one-factor structure.³⁷ These supplementary analyses indicated high ordinal reliability and a broad unidimensional structure, supporting the summed-score approach used here (Supplementary Table 2). Nonetheless, the brief binary format may still have attenuated associations involving SMCs, and future research should consider using more comprehensive SMC measures, ordinal response formats, or reliability coefficients better suited to categorical indicators.^{67,68}

In conclusion, this study provides evidence that loneliness is associated with SMCs partly through depressive symptoms among middle-aged and older Korean adults, and that MIL may attenuate this pathway by buffering the emotional impact of loneliness. Although the moderating effect was modest in magnitude, it was broadly consistent with theoretical expectations and prior empirical findings. The age-stratified analyses also provided preliminary evidence that the pattern of associations among loneliness, depression, and SMCs may differ across life stages, suggesting the need for developmentally informed approaches in future research. Collectively, these findings indicate that interventions addressing both emotion-

al distress and meaningful engagement may hold promise for supporting cognitive well-being in aging populations. Future longitudinal research is needed to clarify the temporal ordering of these associations and to determine whether enhancing MIL can reduce the cognitive consequences of loneliness over time.

Supplementary Materials

The Supplement is available with this article at <https://doi.org/10.30773/pi.2025.0326>.

Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author on reasonable request.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Soo Rim Noh, Sungkun Cho. Data curation: Seung Chul Lee, Dae-Hyun Yun. Formal analysis: Seung Chul Lee, Yerin Shim, Soo Rim Noh. Funding acquisition: Soo Rim Noh, Sungkun Cho. Investigation: Seung Chul Lee, Taehyun Kim. Methodology: Soo Rim Noh, Sungkun Cho, Yerin Shim. Project administration: Soo Rim Noh. Resources: Soo Rim Noh, Taehyun Kim. Software: Seung Chul Lee. Supervision: Soo Rim Noh, Sungkun Cho. Validation: Sungkun Cho, Yerin Shim, Dae-Hyun Yun. Visualization: Seung Chul Lee. Writing—original draft: Seung Chul Lee, Soo Rim Noh. Writing—review & editing: all authors.

ORCID iDs

Seung Chul Lee	https://orcid.org/0009-0003-8505-6934
Sungkun Cho	https://orcid.org/0000-0003-3085-0658
Yerin Shim	https://orcid.org/0000-0002-4976-0882
Dae-Hyun Yun	https://orcid.org/0000-0001-9679-1713
Taehyun Kim	https://orcid.org/0000-0002-6393-8026
Soo Rim Noh	https://orcid.org/0000-0003-2820-7591

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